

Deploying MySQL database using AWS RDS and make a Replica

28/7/2k26

Search Aurora and RDS ->Databases -> Create Database

1. MySQL (all the sql feature)
2. Aurora My(additional features along with Mysql)
3. You can click “i” to know more about the options

Create database [info](#)

Free plan has access to limited features and resources. The free plan limits the features and resources that are available for RDS and Aurora databases. Upgrade your account plan to remove all limitations. [Learn more](#)

[Upgrade plan](#)

Engine options

Engine type [info](#)

☒ Aurora (MySQL Compatible) ☐ Aurora (PostgreSQL Compatible) ☒ MySQL ☐ PostgreSQL

☐ MariaDB ☐ Oracle ☐ Microsoft SQL Server ☐ IBM Db2

Choose a database creation method

☒ Full configuration
 You set all of the configuration options, including ones for availability, security, backups, and maintenance.

☐ Easy create
 Use recommended best-practice configurations. Some configuration options can be changed after the database is created.

Single-AZ DB instance deployment

- it supports 1 instance and no data redundancy
- Both Read and write managed by same instance . so traffic load will be redirect to same server



Select MySQL version and give name (db instance identifier) .

Settings

Edition

☒ MySQL Community

Engine version [info](#)

View the engine versions that support the following database features.

Hide filters

☒ Show only versions that support the Multi-AZ DB cluster [info](#)
Create a Multi-AZ DB cluster with one primary DB instance and two readable standby DB instances. Multi-AZ clusters provide up to 2x faster transaction commit latency and automatic failover in typically under 35 seconds.

☒ Show only versions that support the Amazon RDS Optimized Writes [info](#)
Amazon RDS Optimized Writes improves write throughput by up to 2x at no additional cost.

Engine version

MySQL 8.4.9

☐ Enable RDS Extended Support [info](#)
Amazon RDS Extended Support is a paid offering. By selecting this option, you consent to being charged for this offering if you are running your database major version past the RDS end of standard support date for that version. Check the end of standard support version in the RDS for MySQL documentation.

DB instance identifier [info](#)

Type a name for your DB instance. The name must be unique across all DB instances owned by your AWS account in the current AWS Region.

database-1

The DB instance identifier is case-insensitive, but is stored as all lowercase (as in "mydbinstance"). Constraints: 1 to 63 alphanumeric characters or hyphens. First character must be a letter. Can't contain two consecutive hyphens. Can't end with a hyphen.

Just like there is a master user in MySQL here also we need to give master user name(**admin**)

Self managed –we will manage the

Credentials Settings

Master username [Info](#)
Type a login ID for the master user of your DB instance.
admin

Credentials management
You can use AWS Secrets Manager or manage your master user credentials.

☐ Managed in AWS Secrets Manager - most secure
RDS generates a password for you and manages it throughout its lifecycle using AWS Secrets Manager.

☒ Self managed
Create your own password or have RDS create a password that you manage.

☐ Auto generate password
Amazon RDS can generate a password for you, or you can specify your own password.

Master password [Info](#)
Minimum constraints: At least 8 printable ASCII characters. Can't contain any of the following symbols: / * @

Confirm master password [Info](#)

Keep the credentials password

Additional credentials settings

Database authentication options [Info](#)

☒ Password authentication
Authenticates using database passwords.

☐ Password and IAM database authentication
Authenticates using the database password and user credentials through AWS IAM users and roles.

☐ Password and Kerberos authentication
Choose a directory in which you want to allow authorized users to authenticate with this DB instance using Kerberos Authentication.

Instance type – the server where our MySQL database will be hosted , the instance managed by aws.

Storage type- type of storage will use

Allocated storage – the space we want to allocate (measured in GB , min-20GB , Max -64 TB).

Aurora and RDS > Databases > Create database

☐ Standard classes (includes m classes)
☐ Memory optimized classes (includes r and x classes)
☒ Burstable classes (includes t classes)

Instance type

db.t3.micro
2 vCPU 1 GiB RAM EBS Bandwidth: Up to 2,085 Mbps Network: Up to 5 Gbps

Storage

Storage type [Info](#)
Provisioned IOPS SSD (io2) storage volumes are now available.
General Purpose SSD (gp2)
Baseline performance determined by volume size

Allocated storage [Info](#)
20 GB
Allocated storage value must be 20 GiB to 6,144 GiB

Additional storage configuration

We don't want to connect the database instance.

Keep default vpc

Connectivity [Info](#)

Compute resource
Choose whether to set up a connection to a compute resource for this database. Setting up a connection will automatically change connectivity settings so that the compute resource can connect to this database.

☒ Don't connect to an EC2 compute resource
Don't set up a connection to a compute resource for this database. You can manually set up a connection to a compute resource later.

☐ Connect to an EC2 compute resource
Set up a connection to an EC2 compute resource for this database.

Virtual private cloud (VPC) [Info](#)
Choose the VPC. The VPC defines the virtual networking environment for this DB instance.

Default VPC (vpc-07119e90b6683d8a0)
0 Subnets, 0 Availability Zones

After a database is created, you can't change its VPC. Only VPCs with a corresponding DB subnet group are listed.

DB subnet group – we can provide subnets where we can deploy our DB (here selecting default)

Public – access type of DB

VPC security Group – just like SGs (here default value)

Availability zone – no preference

Then create Database

DB subnet group [Info](#)
Choose the DB subnet group. The DB subnet group defines which subnets and IP ranges the DB instance can use in the VPC that you selected.

default

Public access [Info](#)

☒ Yes
RDS assigns a public IP address to the database. Amazon EC2 instances and other resources outside of the VPC can connect to your database. Resources inside the VPC can also connect to the database. Choose one or more VPC security groups that specify which resources can connect to the database.

☐ No
RDS doesn't assign a public IP address to the database. Only Amazon EC2 instances and other resources inside the VPC can connect to your database. Choose one or more VPC security groups that specify which resources can connect to the database.

VPC security group (firewall) [Info](#)
Choose one or more VPC security groups to allow access to your database. Make sure that the security group rules allow the appropriate incoming traffic.

☒ Choose existing
Choose existing VPC security groups

☐ Create new
Create new VPC security group

Existing VPC security groups
Choose one or more options

default

RDS database-1 instance deployed , its managed by RDS AWS.

Note * -If you want its full control (OS level) then install MySQL in on an instance

Successfully created database database-1 [View connection details](#)

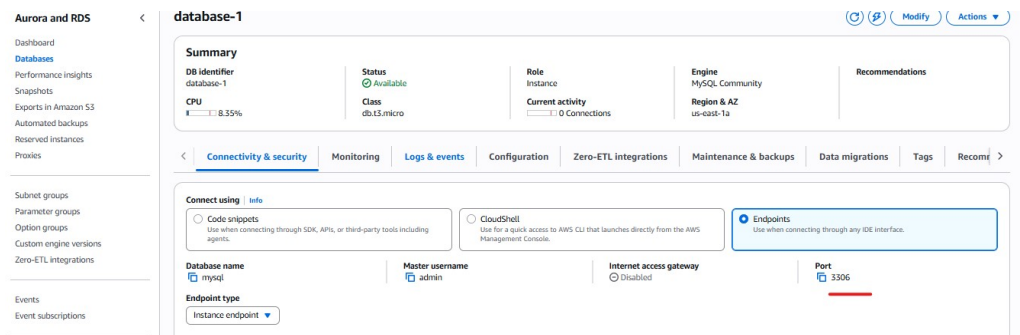
You can use settings from database-1 to simplify configuration of suggested database add-ons while we finish creating your DB for you.

Databases (1) [Group resources](#) [Modify](#) [Actions](#) [Create database](#)

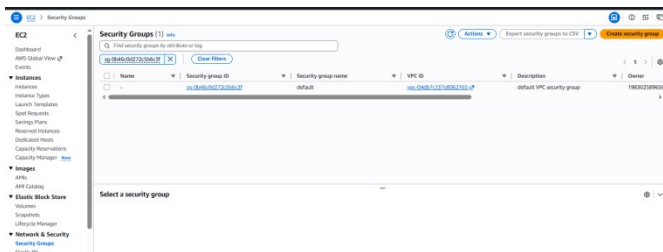
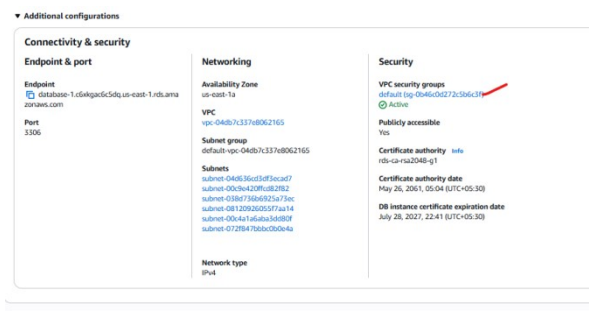
Filter by databases

DB identifier	Status	Role	Engine	Upgrade rollout order	Region ...	Size	Recommendations
database-1	Available	Instance	MySQL Co...	SECOND	us-east-1a	db.t3.micro	

Now configure endpoint for Mysql connect .

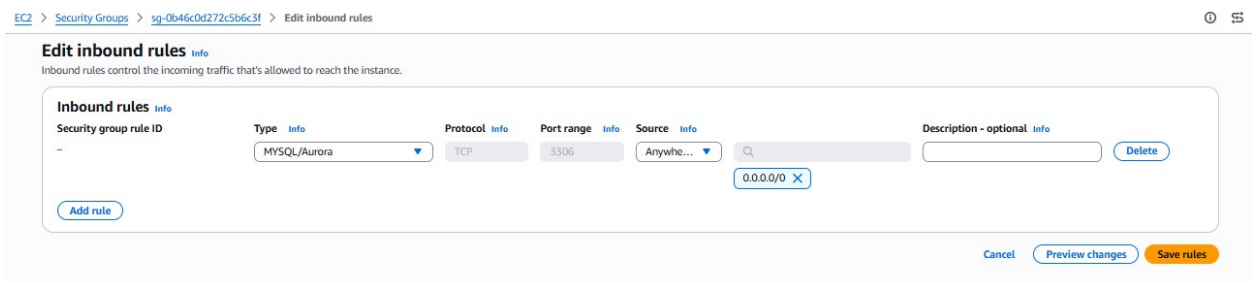


Select security group and allow access for MySQL (port 3306).



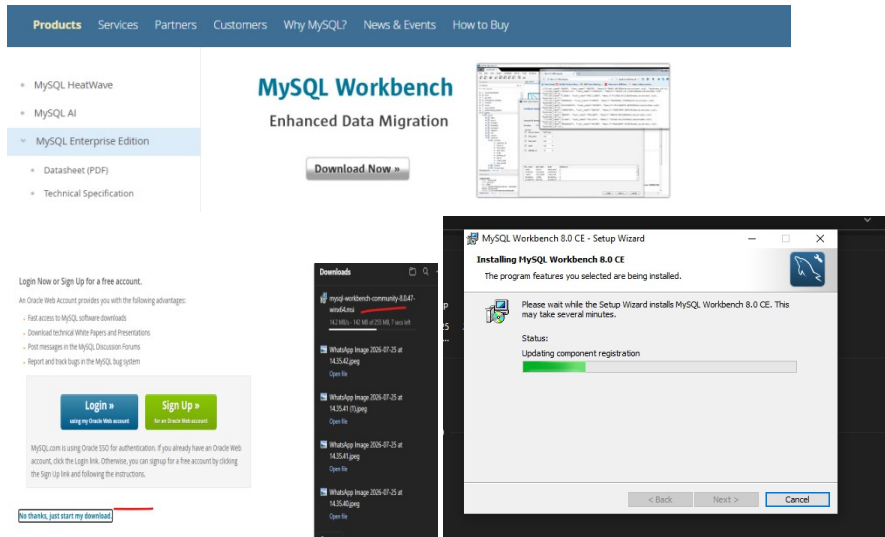
Give inbound rule MSSQL/Aurora (3306) – anywhere.

You can also give all traffic for checking purpose.

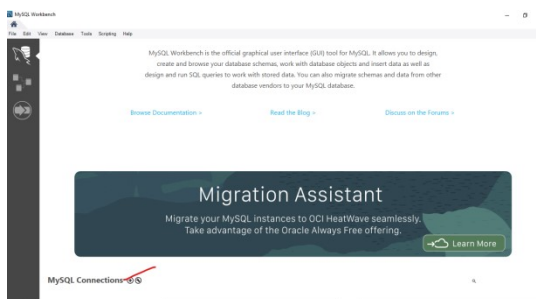


We use Putty ,RDP to connect EC2 instance , like that we will use Mysql Workbench to connect with database-1.

1. Download My Sql workbench .([MySQL :: MySQL Workbench](#))

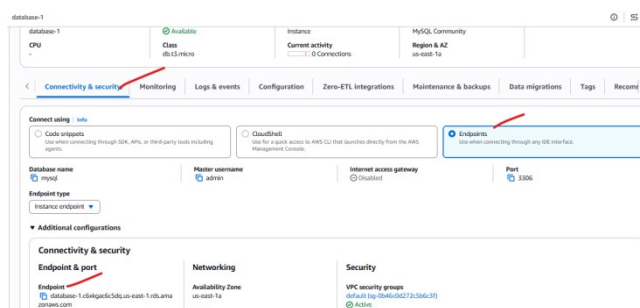


Create a new connection



Connection – give any database name

Hostname – DB instance ip or endpoint (**here endpoint - database-1.c6xkgac6c5dq.us-east-1.rds.amazonaws.com**)



Port – MySql port number (3306)

Username- give the Master username that you had created at DB creation time (admin).

Credentials Settings

Master username [info](#)
Type a login ID for the master user of your DB instances.
admin

Credentials management
You can use AWS Secrets Manager to manage your master user credentials.

☐ Managed in AWS Secrets Manager - most secure
AWS generates a password for you and manages it throughout its lifecycle using AWS Secrets Manager.

☒ Self managed
Create your own password. [View AWS create a password that you manage.](#)

☐ Auto generate password
Amazon RDS can generate a password for you, or you can specify your own password.

Master password [info](#)

Confirm master password [info](#)

Password – give the password for Master username (admin) and click test connection.

MySQL Workbench is the official graphical user interface (GUI) tool for MySQL. It allows you to design, develop, and administer MySQL databases.

Setup New Connection

Connection Name: database-1

Connection Method: Standard (TCP/IP)

Parameters: SSL Advanced

Hostname: database-1.c6vkgac6c5dq.us-east-1.rds.amazonaws.com

Port: 3306

Username: admin

Password: Store in Vault... Clear

Default Schema:

Please enter password for the following service:

Service: Mysql@database-1.c6vkgac6c5dq.us-east-1.rds.amazonaws.com

User: admin

Password:

Test Connection

You will get successful connection . then click ok.

Setup New Connection

Connection Name: database-1

Connection Method: Standard (TCP/IP)

Parameters: SSL Advanced

Hostname: database-1.c6vkgac6c5dq.us-east-1.rds.amazonaws.com

Port: 3306

Username: admin

Password: Store in Vault... Clear

Default Schema:

Test Connection

MySQL Workbench

Successfully made the MySQL connection

Information related to this connection:

Host: database-1.c6vkgac6c5dq.us-east-1.rds.amazonaws.com

Port: 3306

User: admin

SSL: enabled with TLS_AES_128_GCM_SHA256

A successful MySQL connection was made with the parameters defined for this connection.

You will find the database-1 . click the database .

Migration Assistant

Migrate your MySQL instances to OCI HeatWave seamlessly.
Take advantage of the Oracle Always Free offering.

Learn More

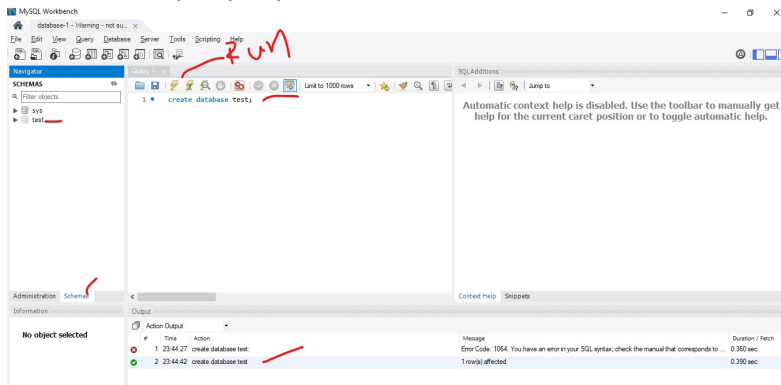
MySQL Connections

database-1

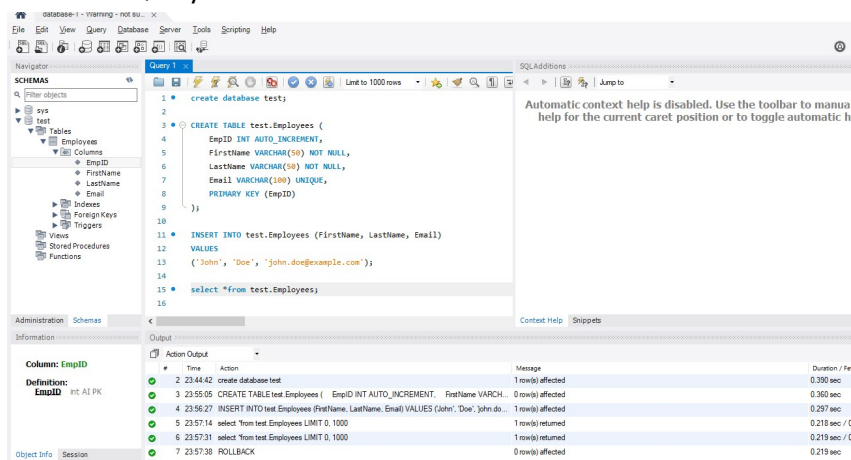
admin

database-1.c6vkgac6c5dq.us-east-1.rds.amazonaws.com

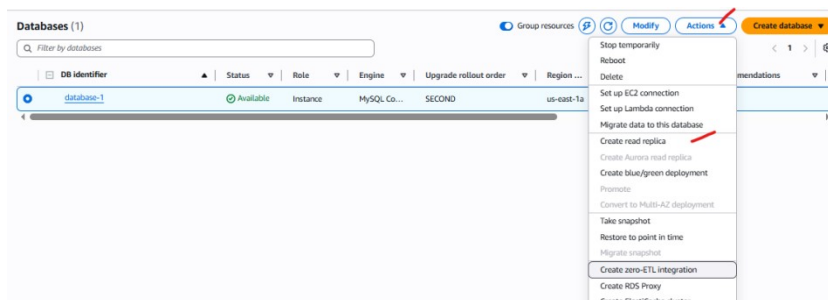
Now run a simple query to create a database named **test**.



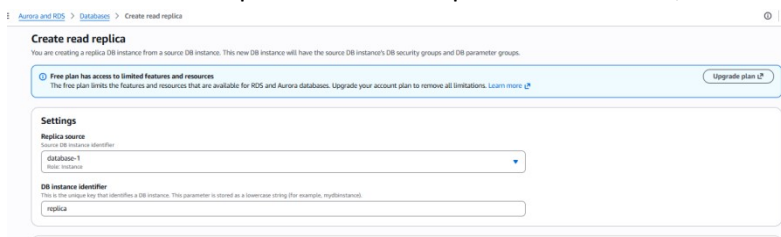
Run some Query



Now create a replica



Give name of the replica . almost all option will be same , until we want some change.



Example: `arn:aws:kms:region::accountID:key=key-id`

Account
198302589656

KMS key ID
c860a09b-1ba2-41f3-91fc-2a055a96ae34

Maintenance
☒ Enable auto minor version upgrade
Enabling auto minor version upgrade will automatically upgrade your database minor version. For limitations and more details, see [Automatically upgrading the minor engine version documentation](#).
☐ Enable deletion protection
Protects the database from being deleted accidentally. While this option is enabled, you can't delete the database.

[Cancel](#) [Create read replica](#)

While creating replica , the primary database-1 also modifying , it means it syncing the data and configuration file in the background by AWS RDS.

Note* - if we use select another region for replica , then it will not show in the primary RDS , we can find the replica in that location only .(select mentioned region > Aurora and RDS)

Creating replica **replica** in US East (N. Virginia)
Your database might take a few minutes to launch.

Databases (2) Group resources Modify Actions Create database

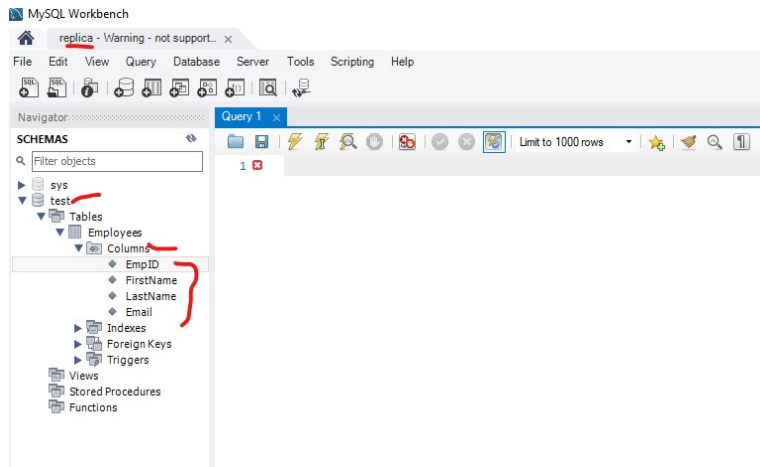
Filter by databases

DB identifier	Status	Role	Engine	Upgrade rollout order	Region ...	Size	Recommendations
database-1	Modify...	Primary	MySQL Co...	SECOND	us-east-1a	db.t3.micro	
replica	Creating	Replica	MySQL Co...	SECOND	-	db.t3.micro	

Once the replica available you can connect the replica in MySQL workbench to check if the replicated data from primary DB is there or not . (user the same process which we followed for primary db)



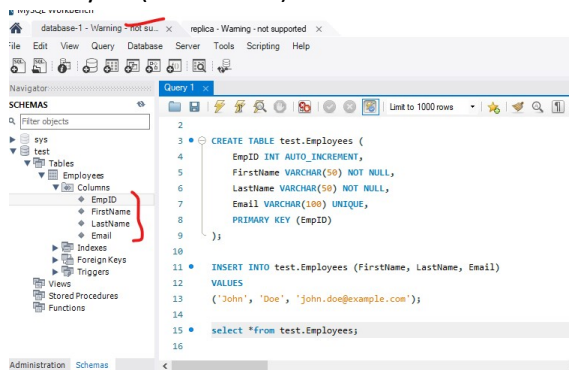
You can observe the database (test) already there in the replica .



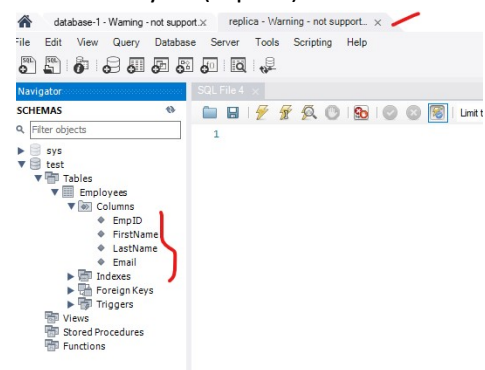
You can also check data sync ,change some data in primary DB and check if it reflecting in the replica.

Before

Primary DB(database-1)



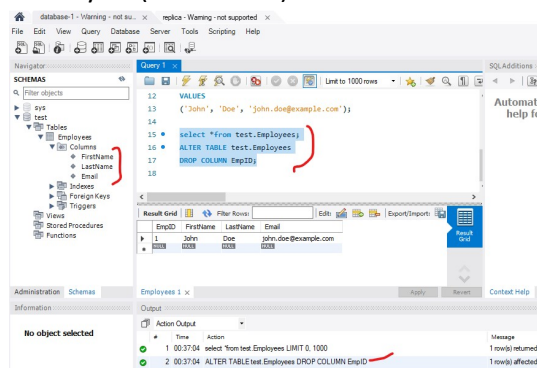
secondary DB(replica)



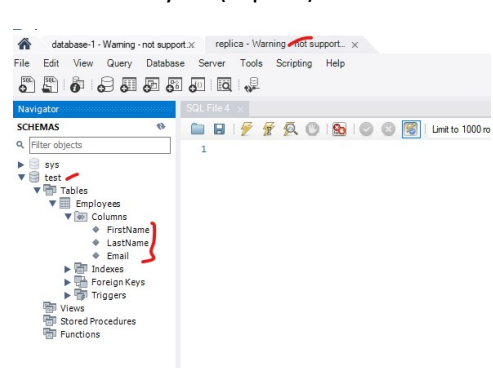
After –

Drop EmpID column in Primary DB(database-1) and see the secondary DB(replica)

Primary DB(database-1)

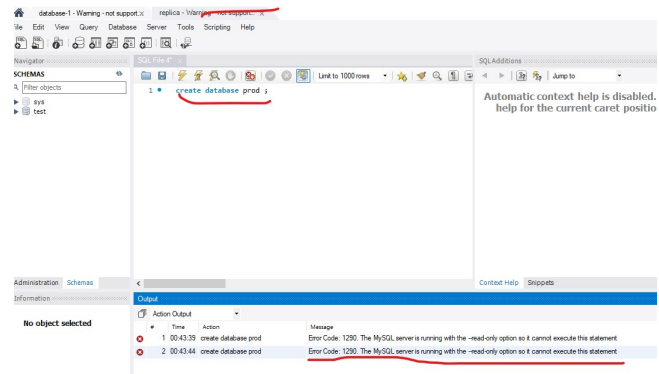


secondary DB(replica)



Now make change in secondary DB(replica) and see if its replicating in Primary DB(database-1)

secondary DB(replica)



Since Replica has read-only permission , so we can not create /write/upload any data in replica database.

Task -completed